

Configuring sFlow

This chapter contains the following sections:

- Information About sFlow, on page 1
- Prerequisites, on page 2
- Guidelines and Limitations for sFlow, on page 2
- Default Settings for sFlow, on page 2
- Configuring sFLow, on page 2
- Verifying the sFlow Configuration, on page 9
- Configuration Examples for sFlow, on page 9
- Additional References for sFlow, on page 9
- Feature History for sFlow, on page 10

Information About sFlow

sFlow allows you to monitor the real-time traffic in data networks that contain switches and routers. It uses the sampling mechanism in the sFlow Agent software on switches and routers for monitoring traffic and to forward the sample data on ingress and egress ports to the central data collector, also called the sFlow Analyzer.

For more information about sFlow, see RFC 3176.

sFlow Agent

The sFlow Agent, which is embedded in the Cisco NX-OS software, periodically samples or polls the interface counters that are associated with a data source of the sampled packets. The data source can be an Ethernet interface, an EtherChannel interface, or a range of Ethernet interfaces. The sFlow Agent queries the Ethernet port manager for the respective EtherChannel membership information and also receives notifications from the Ethernet port manager for membership changes.

When you enable sFlow sampling in the Cisco NX-OS software, based on the sampling rate and the hardware internal random number, the ingress packets and egress packets are sent to the CPU as an sFlow-sampled packet. The sFlow Agent processes the sampled packets and sends an sFlow datagram to the sFlow Analyzer. In addition to the original sampled packet, an sFlow datagram includes the information about the ingress port, egress port, and the original packet length. An sFlow datagram can have multiple sFlow samples.

Prerequisites

You must enable the sFlow feature using the **feature sflow** command to configure sFlow.

Guidelines and Limitations for sFlow

The sFlow configuration guidelines and limitations are as follows:

- When you enable sFlow for an interface, it is enabled for both ingress and egress. You cannot enable sFlow for only ingress or only egress.
- sFlow egress sampling for multicast, broadcast, or unknown unicast packets is not supported.
- You should configure the sampling rate based on the sFlow configuration and traffic in the system.
- Cisco Nexus 3000 Series supports only one sFlow collector.

Default Settings for sFlow

Table 1: Default sFlow Parameters

Parameters	Default
sFlow sampling-rate	4096
sFlow sampling-size	128
sFlow max datagram-size	1400
sFlow collector-port	6343
sFlow counter-poll-interval	20

Configuring sFLow

Enabling the sFlow Feature

You must enable the sFlow feature before you can configure sFlow on the switch.

Procedure

	Command or Action	Purpose
Step 1	switch# **configure terminal**	Enters global configuration mode.
Step 2	[**no**] **feature sflow**	Enables the sFlow feature.

	Command or Action	Purpose
Step 3	(Optional) **show feature**	Displays enabled and disabled features.
Step 4	(Optional) switch(config)# **copy running-config startup-config**	Saves the change persistently through reboots and restarts by copying the running configuration to the startup configuration.

Example

The following example shows how to enable the sFlow feature:

```
switch# configure terminal
switch(config)# feature sflow
switch(config)# copy running-config startup-config
```

Configuring the Sampling Rate

Before you begin

Ensure that you have enabled the sFlow feature.

Procedure

	Command or Action	Purpose
Step 1	switch# **configure terminal**	Enters global configuration mode.
Step 2	[**no**] **sflow sampling-rate** *sampling-rate*	Configures the sFlow sampling rate for packets. The *sampling-rate* can be an integer between 4096-1000000000. The default value is 4096.
Step 3	(Optional) **show sflow**	Displays sFlow information.
Step 4	(Optional) switch(config)# **copy running-config startup-config**	Saves the change persistently through reboots and restarts by copying the running configuration to the startup configuration.

Example

This example shows how to set the sampling rate to 50,000:

```
switch# configure terminal
switch(config)# sflow sampling-rate 50000
switch(config)# copy running-config startup-config
```

Configuring the Maximum Sampled Size

You can configure the maximum number of bytes that should be copied from a sampled packet.

Before you begin

Ensure that you have enabled the sFlow feature.

Procedure

	Command or Action	Purpose
Step 1	switch# **configure terminal**	Enters global configuration mode.
Step 2	[**no**] **sflow max-sampled-size** *sampling-size*	Configures the sFlow maximum sampling size packets. The range for the *sampling-size* is from 64 to 256 bytes. The default value is 128.
Step 3	(Optional) **show sflow**	Displays sFlow information.
Step 4	(Optional) switch(config)# **copy running-config startup-config**	Saves the change persistently through reboots and restarts by copying the running configuration to the startup configuration.

Example

This example shows how to configure the maximum sampling size for the sFlow Agent:

```
switch# configure terminal
switch(config)# sflow max-sampled-size 200
switch(config)# copy running-config startup-config
```

Configuring the Counter Poll Interval

You can configure the maximum number of seconds between successive samples of the counters that are associated with the data source. A sampling interval of 0 disables counter sampling.

Before you begin

Ensure that you have enabled the sFlow feature.

Procedure

	Command or Action	Purpose
Step 1	switch# **configure terminal**	Enters global configuration mode.
Step 2	[**no**] **sflow counter-poll-interval** *poll-interval*	Configures the sFlow poll interval for an interface. The range for the *poll-interval* is from 0 to 2147483647 seconds. The default value is 20.
Step 3	(Optional) **show sflow**	Displays sFlow information.

	Command or Action	Purpose
Step 4	(Optional) switch(config)# **copy running-config startup-config**	Saves the change persistently through reboots and restarts by copying the running configuration to the startup configuration.

Example

This example shows how to configure the sFlow poll interval for an interface:

```
switch# configure terminal
switch(config)# sflow counter-poll-interval 100
switch(config)# copy running-config startup-config
```

Configuring the Maximum Datagram Size

You can configure the maximum number of data bytes that can be sent in a single sample datagram.

Before you begin

Ensure that you have enabled the sFlow feature.

Procedure

	Command or Action	Purpose
Step 1	switch# **configure terminal**	Enters global configuration mode.
Step 2	[**no**] **sflow max-datagram-size** *datagram-size*	Configures the sFlow maximum datagram size. The range for the *datagram-size* is from 200 to 9000 bytes. The default value is 1400.
Step 3	(Optional) **show sflow**	Displays sFlow information.
Step 4	(Optional) switch(config)# **copy running-config startup-config**	Saves the change persistently through reboots and restarts by copying the running configuration to the startup configuration.

Example

This example shows how to configure the sFlow maximum datagram size:

```
switch# configure terminal
switch(config)# sflow max-datagram-size 2000
switch(config)# copy running-config startup-config
[########################################] 100%
```

Configuring the sFlow Analyzer Address

Before you begin

Ensure that you have enabled the sFlow feature.

Procedure

	Command or Action	Purpose
Step 1	switch# **configure terminal**	Enters global configuration mode.
Step 2	[**no**] **sflow collector-ip** *IP-address vrf-instance*	Configures the IPv4 address for the sFlow Analyzer. *vrf-instance* can be one of the following: • A user-defined VRF name—You can specify a maximum of 32 alphanumeric characters. • **vrf management**— You must use this option if the sFlow data collector is on the network connected to the management port. • **vrf default**— You must use this option if the sFlow data collector is on the network connected to the front panel ports.
Step 3	(Optional) **show sflow**	Displays sFlow information.
Step 4	(Optional) switch(config)# **copy running-config startup-config**	Saves the change persistently through reboots and restarts by copying the running configuration to the startup configuration.

Example

This example shows how to configure the IPv4 address of the sFlow data collector that is connected to the management port:

```
switch# configure terminal
switch(config)# sflow collector-ip 192.0.2.5 vrf management
switch(config)# copy running-config startup-config
```

Configuring the sFlow Analyzer Port

You can configure the destination port for sFlow datagrams.

Before you begin

Ensure that you have enabled the sFlow feature.

Procedure

	Command or Action	Purpose
Step 1	switch# **configure terminal**	Enters global configuration mode.
Step 2	[**no**] **sflow collector-port** *collector-port*	Configures the UDP port of the sFlow Analyzer. The range for the *collector-port* is from 0 to 65535. The default value is 6343.
Step 3	(Optional) **show sflow**	Displays sFlow information.
Step 4	(Optional) switch(config)# **copy running-config startup-config**	Saves the change persistently through reboots and restarts by copying the running configuration to the startup configuration.

Example

This example shows how to configure the destination port for sFlow datagrams:

```
switch# configure terminal
switch(config)# sflow collector-port 7000
switch(config)# copy running-config startup-config
[########################################] 100%
switch(config)#
```

Configuring the sFlow Agent Address

Before you begin

Ensure that you have enabled the sFlow feature.

Procedure

	Command or Action	Purpose
Step 1	switch# **configure terminal**	Enters global configuration mode.
Step 2	[**no**] **sflow agent-ip** *ip-address*	Configures the IPv4 address of the sFlow Agent. The default *ip-address* is 0.0.0.0, which means that all sampling is disabled on the switch. You must specify a valid IP address to enable sFlow functionality. **Note** This IP address is not necessarily the source IP address for sending the sFlow datagram to the collector.

	Command or Action	Purpose
Step 3	(Optional) **show sflow**	Displays sFlow information.
Step 4	(Optional) switch(config)# **copy running-config startup-config**	Saves the change persistently through reboots and restarts by copying the running configuration to the startup configuration.

Example

This example shows how to configure the IPv4 address of the sFlow Agent:

```
switch# configure terminal
switch(config)# sflow agent-ip 192.0.2.3
switch(config)# copy running-config startup-config
```

Configuring the sFlow Sampling Data Source

The sFlow sampling data source can be an Ethernet port, a range of Ethernet ports, or a port channel.

Before you begin

- Ensure that you have enabled the sFlow feature.
- If you want to use a port channel as the data source, ensure that you have already configured the port channel and you know the port channel number.

Procedure

	Command or Action	Purpose
Step 1	switch# **configure terminal**	Enters global configuration mode.
Step 2	switch(config)# [**no**] **sflow data-source interface** [**ethernet** *slot*/*port*[*-port*] |**port-channel** *channel-number*]	Configures the sFlow sampling data source. For an Ethernet data source, *slot* is the slot number and *port* can be either a single port number or a range of ports designated as *port-port*.
Step 3	(Optional) switch(config)# **show sflow**	Displays sFlow information.
Step 4	(Optional) switch(config)# **copy running-config startup-config**	Saves the change persistently through reboots and restarts by copying the running configuration to the startup configuration.

Example

This example shows how to configure Ethernet ports 5 through 12 for the sFlow sampler:

```
switch# configure terminal
switch(config)# sflow data-source interface ethernet 1/5-12
switch(config)# copy running-config startup-config
[########################################] 100%
switch(config)#
```

This example shows how to configure port channel 100 for the sFlow sampler:

```
switch# configure terminal
switch(config)# sflow data-source interface port-channel 100
switch(config)# copy running-config startup-config
[########################################] 100%
switch(config)#
```

Verifying the sFlow Configuration

Use the following commands to verify the sFlow configuration information:

Command	Purpose
show sflow	Displays the sFlow global configuration.
show sflow statistics	Displays the sFlow statistics.
clear sflow statistics	Clears the sFlow statistics.
show running-config sflow [**all**]	Displays the current running sFlow configuration.

Configuration Examples for sFlow

This example shows how to configure sFlow:

```
feature sflow
sflow sampling-rate 5000
sflow max-sampled-size 200
sflow counter-poll-interval 100
sflow max-datagram-size 2000
sflow collector-ip 192.0.2.5 vrf management
sflow collector-port 7000
sflow agent-ip 192.0.2.3
sflow data-source interface ethernet 1/5
```

Additional References for sFlow

Table 2: Related Documents for sFlow

Related Topic	Document Title
sFlow CLI commands	*Cisco Nexus 3000 Series NX-OS System Management Command Reference.*

Related Topic	Document Title
RFC 3176	Defines the sFlow packet format and SNMP MIB. http://www.sflow.org/rfc3176.txt

Feature History for sFlow

This table includes only the updates for those releases that have resulted in additions or changes to the feature.

Feature Name	Releases	Feature Information
sFlow	5.0(3)U4(1)	This feature was introduced.